Internal Capital Frictions in Intermediaries and Exchange Rate Dynamics

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Can the Convergence Risk and Internal Capital Frictions Explain FX Arbitrage Deviations?

Why CIP Arbitrage Is Actually Risky: The Convergence Risk

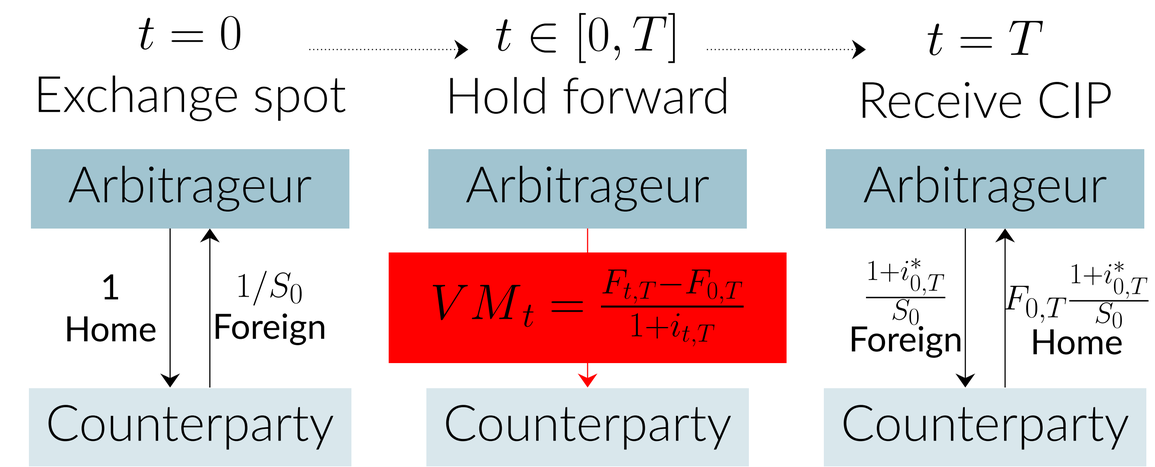
Significant changes in the mark-to-market values of outstanding derivative contracts of large dealers lead to sizable variation margins. These internal liquidity shocks prevent granular intermediaries from arbitraging CIP deviations away because of the marginal value of cash.

Channel: Variation Margins on FX Forwards Create Liquidity Risk

Covered Interest Rate Parity (CIP): $F_{0,T} = S_0 \frac{1+i_{0,T}-b_T}{1+i^*_{0,T}}$, with deviation b_T.

The puzzle: CIP bases $b_{c,T}$ are **not** equalized across currencies. Why?

Our hypothesis: variation margins VM_t on the FX forward leg of the arbitrage create **convergence risk**, mark-to-market losses as prices move.



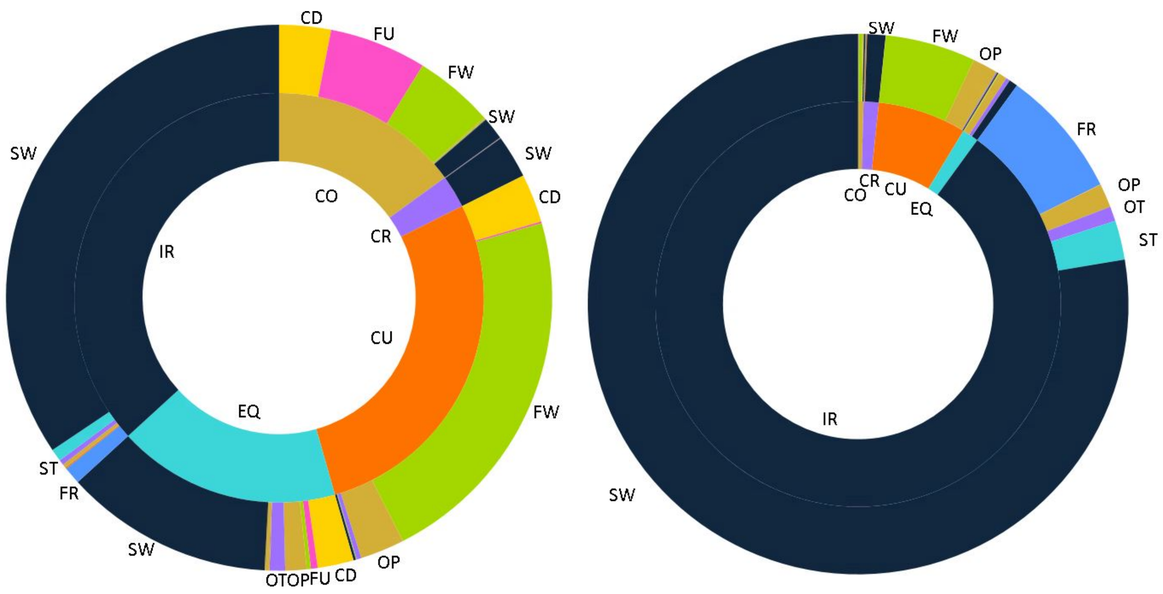
The drivers of FX forwards VMs can indeed explain changes in CIP bases:

$\Delta b_{c,t}$	$\Delta Spot_{c,t}$	$\Delta i_{c,t}$	$\Delta i_{USD,t}$	$\Delta IVol_{c,t}$
Coefficient	0.0231	-0.0417**	0.0905***	-0.0886**
Std. Error	(0.0901)	(0.0132)	(0.0156)	(0.0285)
Observations: 26,512		Number of currencies c: 9		R^2: 0.013

Table: CIP changes regressed on changes in bilateral spot rate, interest rates, and implied volatility.
Notes: Daily changes. Robust standard errors in parentheses, two-way clustered by currency and date.

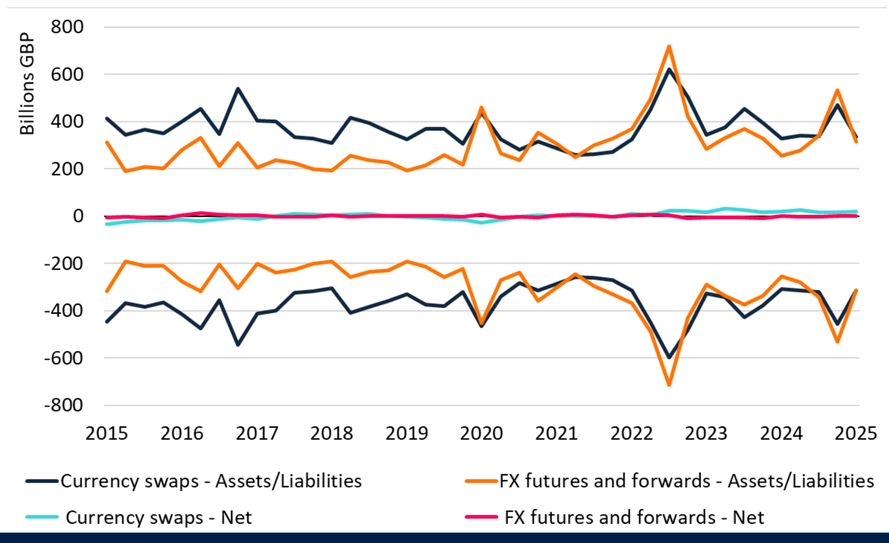
(UK-)EMIR Data: A Unique View Into All Outstanding Derivatives

We access **all derivative trades** with a UK entity at the Bank of England:



FX forwards: 4 million transactions outstanding, USD 40 trillion total notional, at least 40% of the global market (FX trading capital: London).

Result 1: UK banks' FX forwards VMs $\simeq$ annual operating profits. Large!



FX Forwards Market Structure: OTC and Highly Intermediated

Result 2: Large, granular dealers intermediate the OTC, bilateral market.

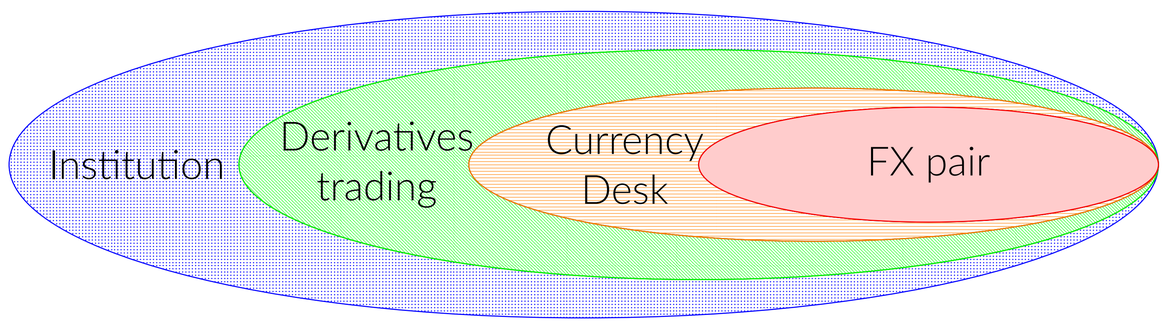
➤ Dealers not substitutable: idiosyncratic shocks matter for CIP bases.



Result 3: Large dispersion in forward rates across dealers and clients.

Testing For Internal Frictions and Balance-Sheet Segmentation

Framework: because of costly external funds and agency costs, banks' treasuries **limit internal liquidity risk** and prevent CIP arbitrage. Use this to map the limit of the firm over dealers balance-sheets' partition $\mathbb{J}$:



Identification hypothesis: CIP is a no-arbitrage condition, so deviations must reflect limited arbitrage capital, which plausibly sits within dealers.

Run **contract-level regression** of CIP basis on dealers' past variation margins, control for high-dimensional fixed effects (including dealer-client):

$$b_\tau = \sum_{j \in \mathbb{J}(c(\tau))} \beta_j \Delta pnl_{d(\tau),j,t(\tau)-1} + \gamma \mathbf{X}_{\tau,d(\tau),c(\tau),t(\tau)} + \varepsilon_\tau$$

What we find: evidence of balance-sheet segmentation, reject the hypothesis of a representative dealer sector, idiosyncratic shocks matter.

Policy implications: variation margins reallocate liquidity across counterparties, but, with granular agents, this is **not** neutral and impact prices.

Literature Review: Where We Fit In

➤ FX Forward / OTC Market Structure and the Granular Hypothesis
➤ Limits to Arbitrage and Intermediary Asset Pricing
➤ Deviations from Covered Interest Parity
➤ Internal Capital Frictions and the Marginal Value of Cash

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